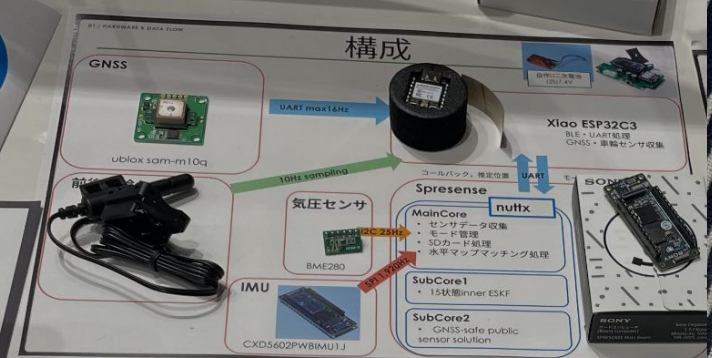


アプリ側
phone app

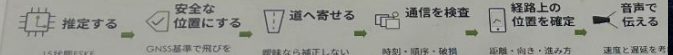


Bluetoothイヤホン
Bluetooth earphone



位置飛びを止めて案内へ

自転車側で「表示してよい位置」を作り、スマホ側で経路上の位置を数回確認してから案内します。一度の位置飛びでは進捗を変えません。



A. 自転車側：案内に渡してよい位置だけを作る
内部推定が一時的に乱れても戻れるGNSS基準を残し、地図補正は内部へ戻しません。公開位置は「安全な基準」に、信頼できる移動量と道路補正だけを重ねて作ります。

品質と位置飛びを確認したGNSSを、ESKFとは別の安全基準として残します。GNSS更新の間はドップラー速度でつなぎ、内部推定が乱れても戻れる位置を保ちます。

1 GNSSを安全基準に
ESKFの絶対位置へ一気に合わせず、前回から「どれだけ動いたか」だけを上限付きで加えます。信用できない時はGNSS基準へ戻します。

2 推定の变化だけ利用
近頃の道路候補が位置・向き・進み方で一致した時だけ、主に道路への横すれを滑らかに補正します。候補が曖昧なら補正しません。

3 道路へ慎重に補正
近頃の道路候補が位置・向き・進み方で一致した時だけ、主に道路への横すれを滑らかに補正します。候補が曖昧なら補正しません。

間隔的な飛び
公開位置は飛びを抑え、地図が曖昧なら無様に寄せない

B. スマホ側：1回の候補ですぐ案内しない
受信後3秒以内の位置から、推定位置→融合前GNSS→スマホGNSSに選びます。距離・向き・これまでの進み方で経路上の候補を絞り。

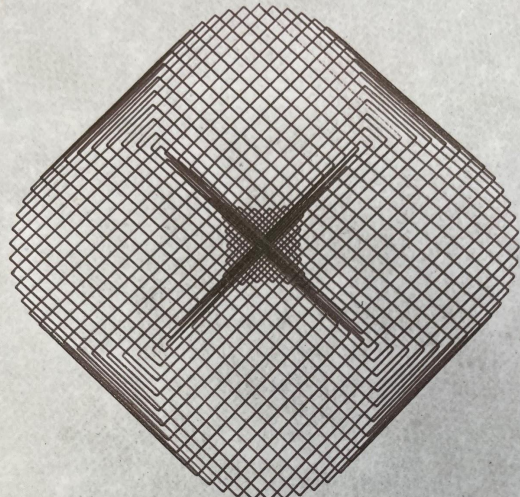
位置の優先順位①推定位置 ②GNSS位置 ③スマホ
経路照合 横ずれ 向き 進み

1回の候補と、3回確認した位置を分ける
大きな前方飛び

確定位置
大きな飛びを検出
候補の飛び幅 > 60m + 速度 × 経路照合

発話時だけ前方補正
前方補正 = min(25m, 速度 × 位置)

自転車側は「表示してよい位置」を作り、スマホ側は「経路上のどこか」を確認して、案内する時刻を決めます。

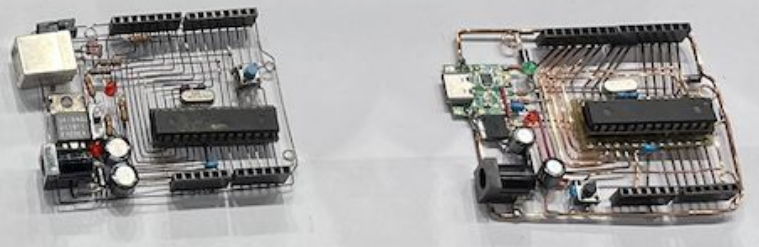
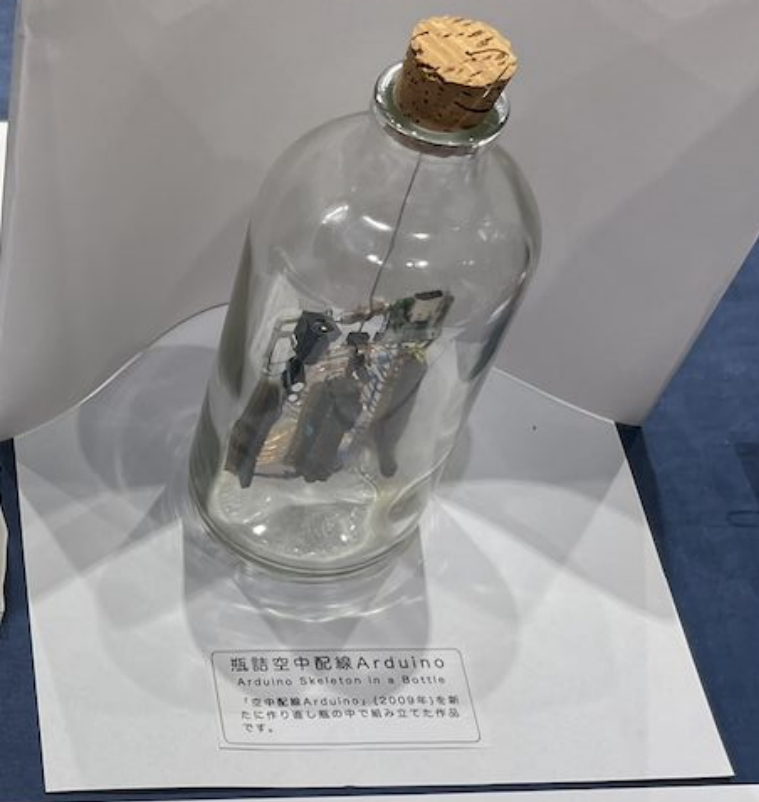


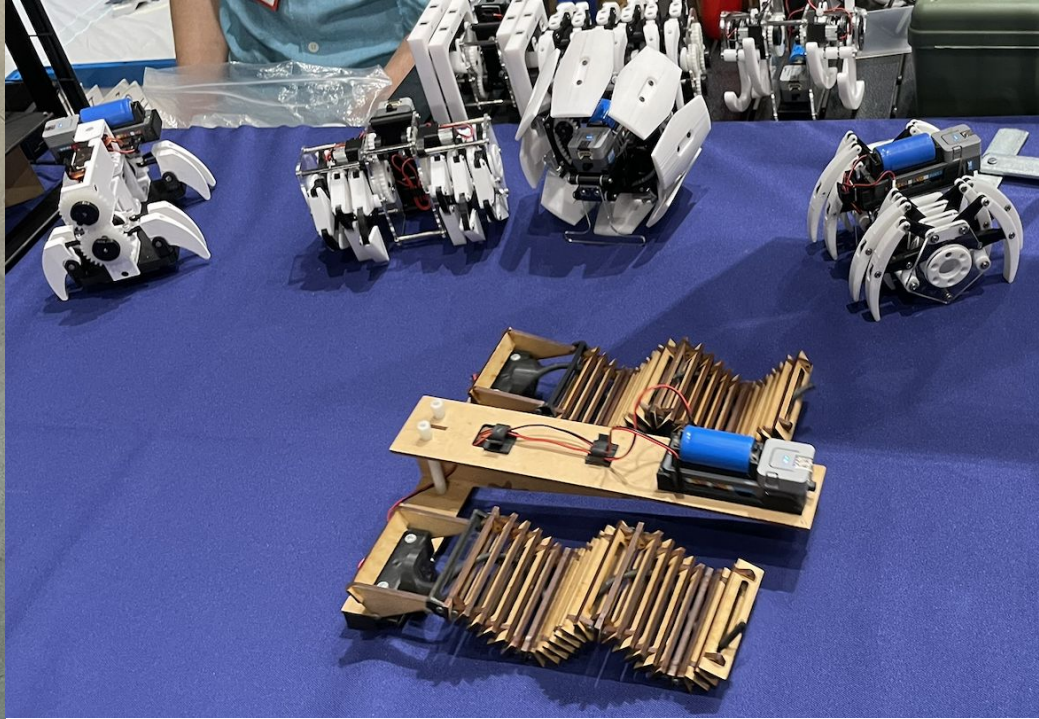
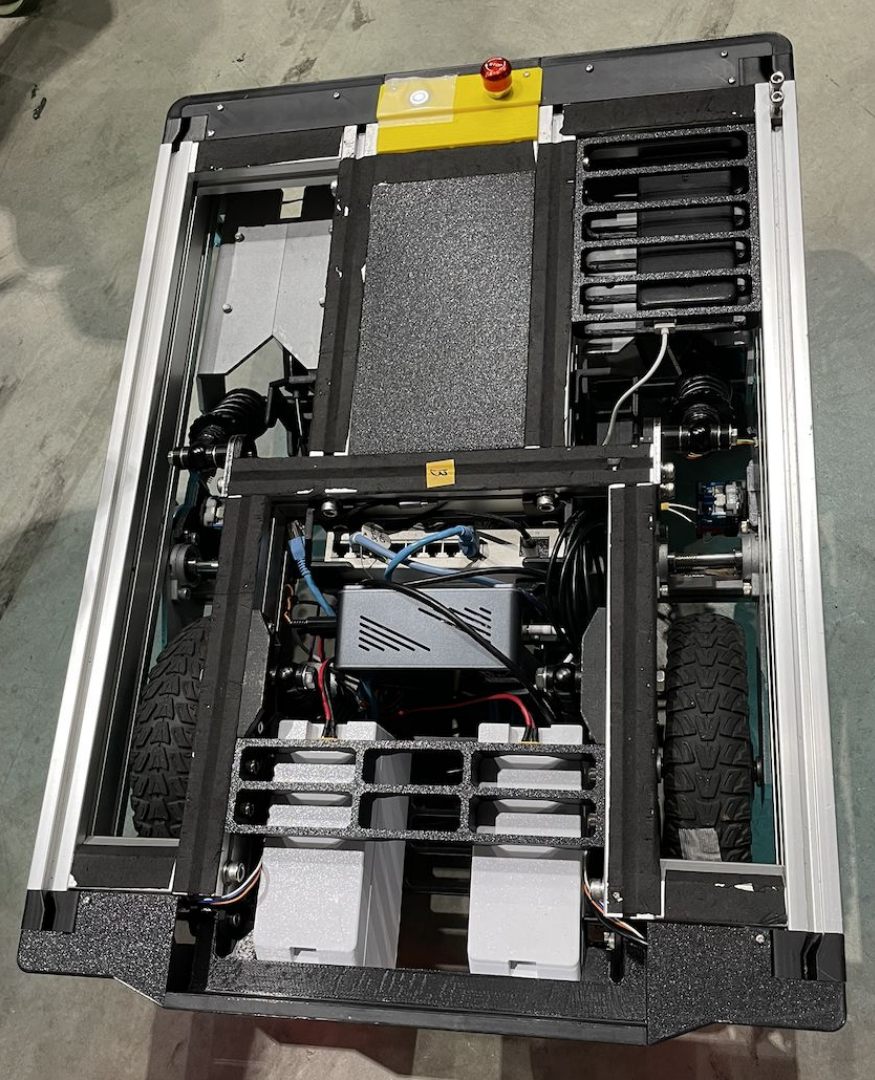
Spiro Maker Mini

Maker Faire Tokyo 2026

Yara's Makers







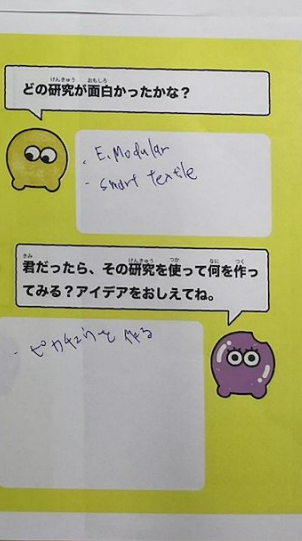
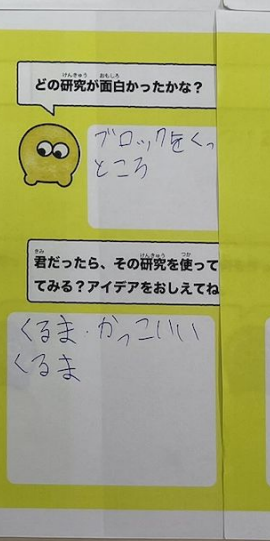
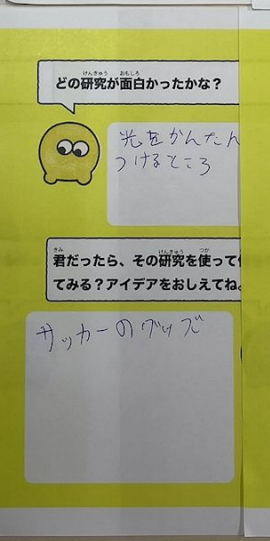
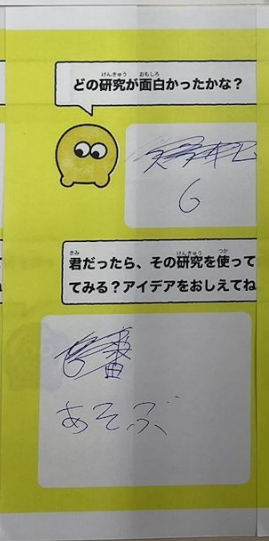
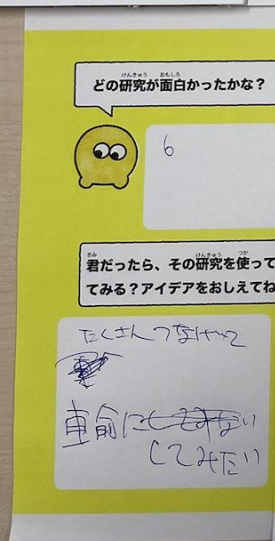
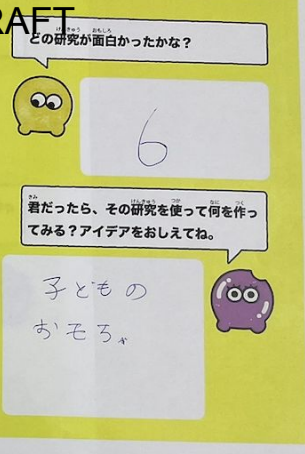
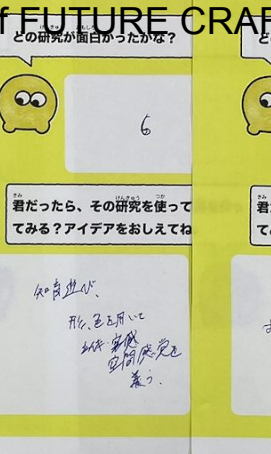
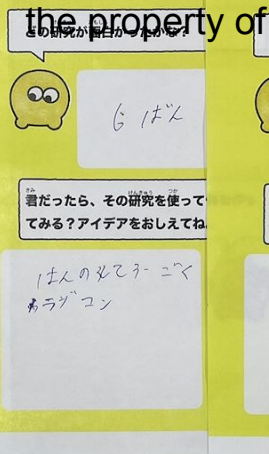
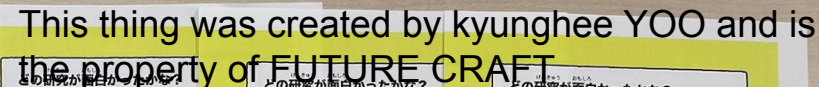
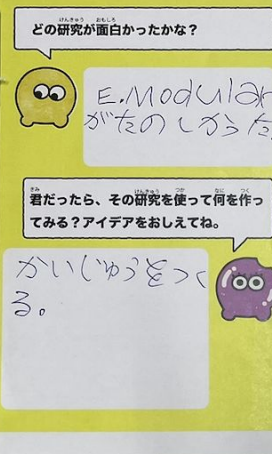
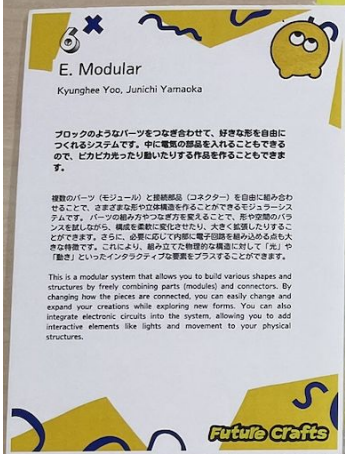






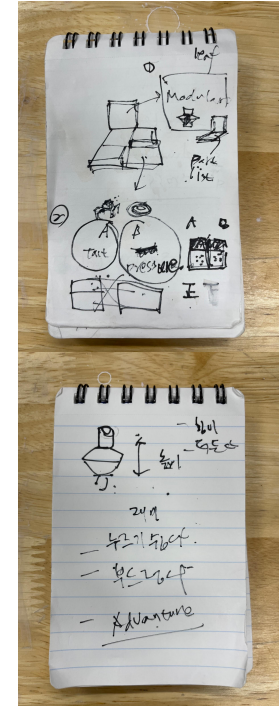
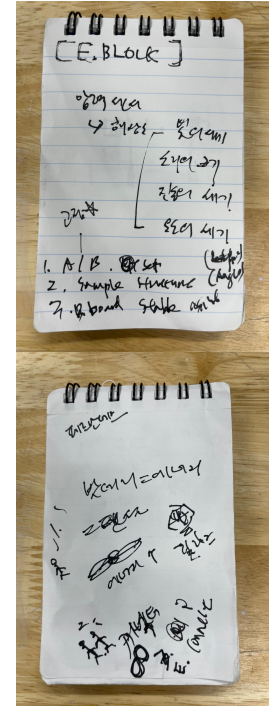
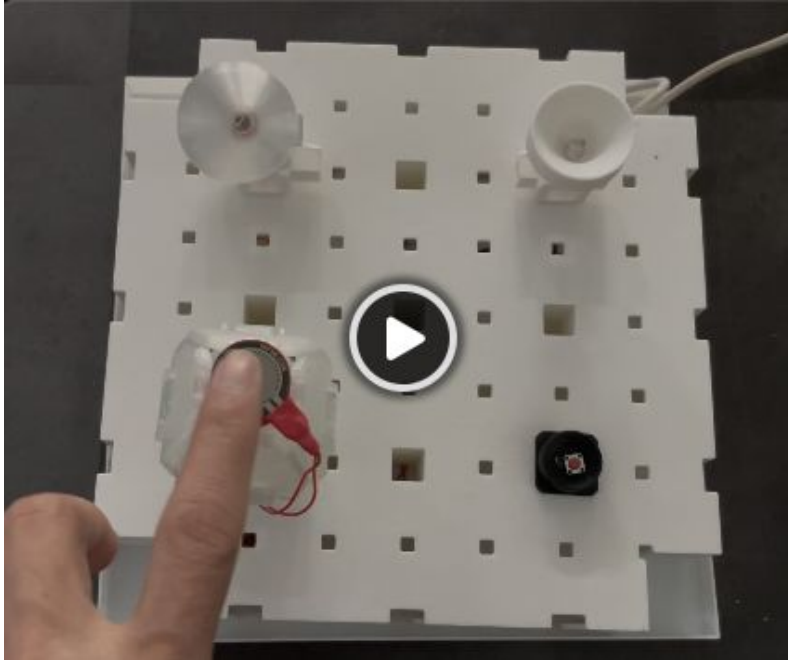
A modular system composed of interchangeable modules and connectors, allowing a wide variety of forms and structures to emerge through flexible combinations. Each component can be assembled, rearranged, and extended in different ways, encouraging experimentation with shape, connection, and spatial relationships. Optional electronic circuits can also be integrated into the structure, adding interactive and functional elements to the physical configuration.

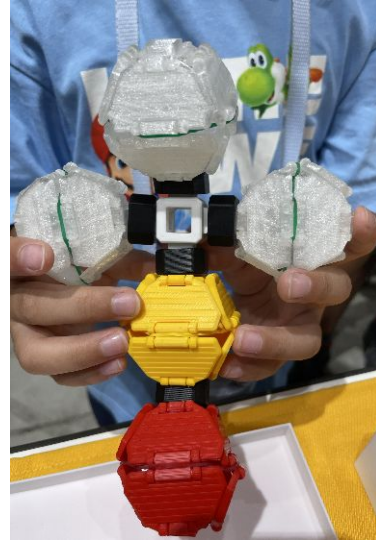
複数のパーツ(モジュール)と接続部品(コネクタ)を自由に組み合わせることで、さまざまな形や構造をつくることができるモジュラシステムです。パーツの組み方やつなぎ方を変えることで、形態や空間の関係を試しながら、柔軟に構成を変化・拡張することができず。また、必要に応じて電子回路を組み込むことも可能で、物理的な構造に光や動きなどのインタラクティブな要素を加えることが可能です。



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Sep 8 - Maker Fair Tokyo Day2







This thing was created by kyunghee YOO and is the property of FUTURE CRAFT

Sep 5 - Maker Fair Tokyo Day1



Sep 4 - Booth Setup Day0

メイカーフェアトウキョウ2026
This thing was created by kyunghee YOQ and is the property of FUTURE CRAFT

Maker Faire Tokyo

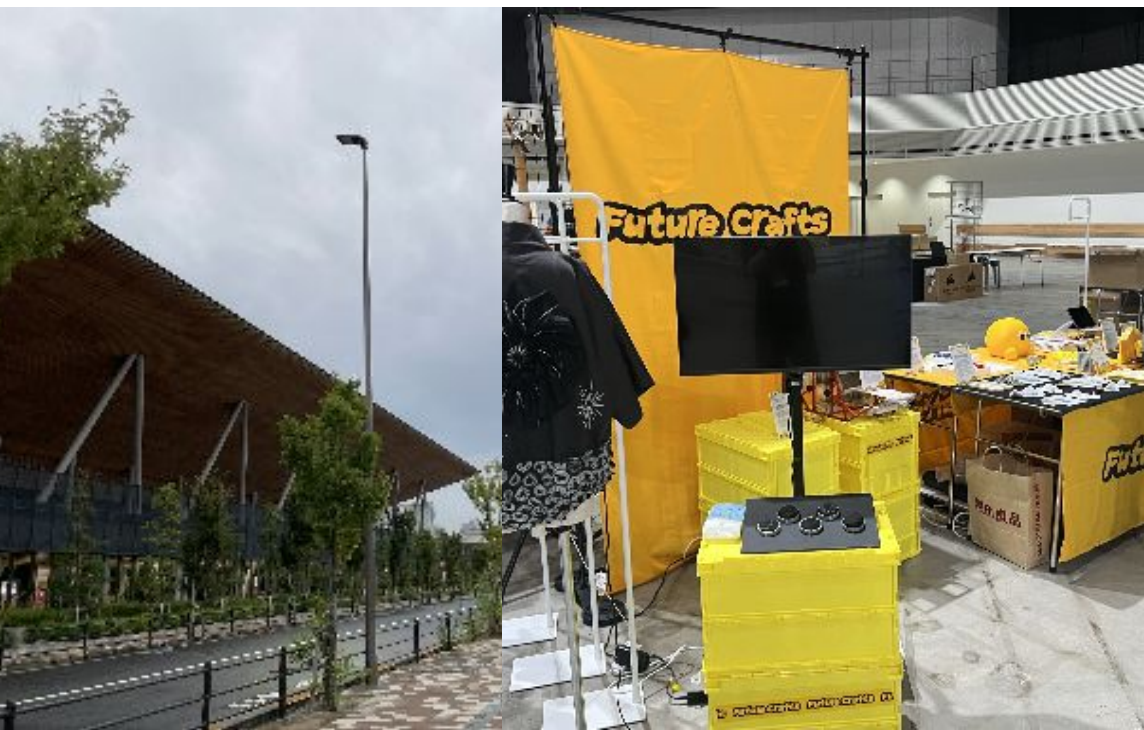
2026年9月
5日(土)12:00-18:00
6日(日)10:00-17:00

MAKE — PLAY — CONNECT — LEARN

有明GYM-EX

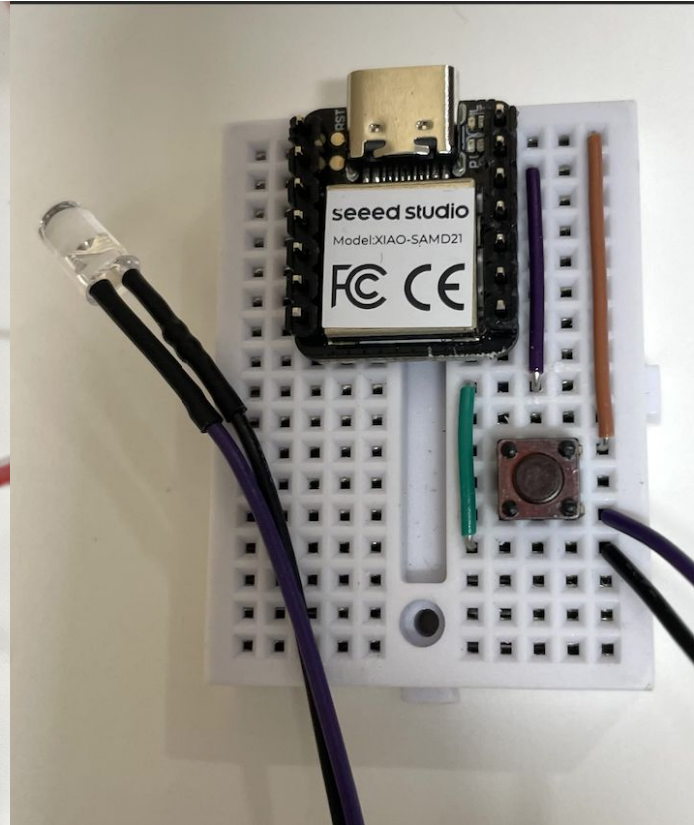
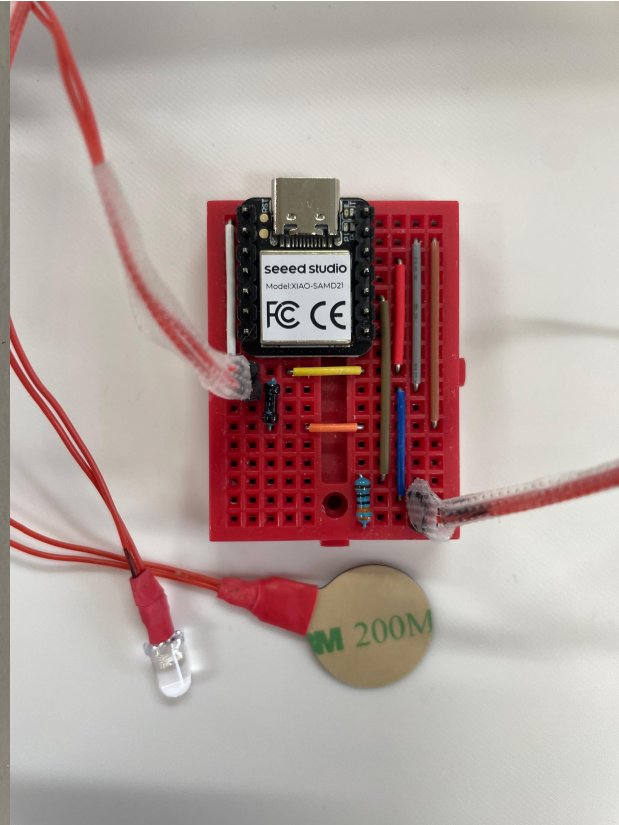
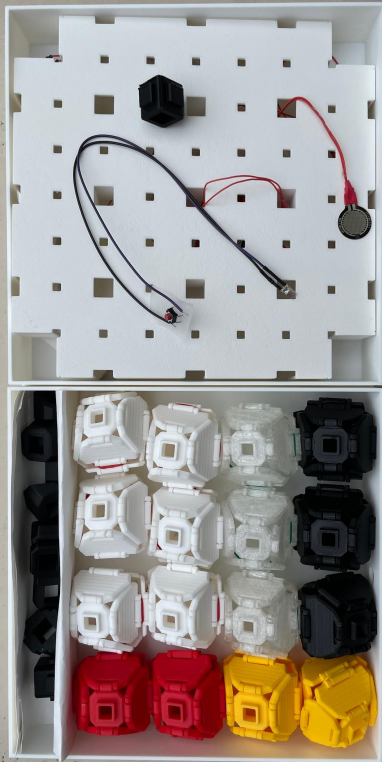
入場料 前売(12/24まで):大人1,000円 | 18歳未満400円 当日:大人1,400円 | 大人(7/24前):1,000円 | 18歳未満500円
当日:大人1,800円 | 大人(7/24前):1,300円 | 18歳未満700円

主催 株式会社インプレス 「入場券は必ず300円以上で」 入場券は必ず300円以上で「入場券は必ず300円以上で」



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Sep 2, 3 - Modular Improvement 4



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Sep 1 - Modular Improvement 3

Arduino code (Plato_001_fsr_led.ino)

```
const int farPin = A0;
const int ledPin = 9;

const float R_FIXED = 1000.0; // 1kΩ
const float VCC = 3.3; // 3.3V 전원

float condMin = 0.00007;
float condMax = 0.001; // 저항 범위

void setup() {
  Serial.begin(9600);
  pinMode(ledPin, OUTPUT);
  pinMode(farPin, INPUT);
}

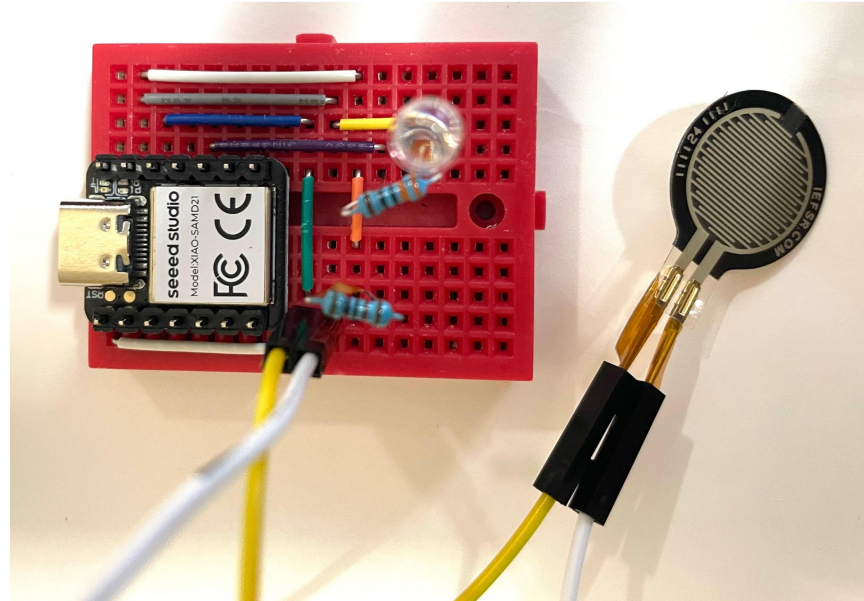
void loop() {
  int raw = analogRead(farPin);
  float vOut = raw * (VCC / 1023.0);

  float conductance;
  if (vOut <= 0.01) {
    conductance = 0;
  } else {
    float rFar = R_FIXED * (VCC - vOut) / vOut;
    conductance = 1.0 / rFar;
  }

  int brightness = (int)((conductance - condMin) / (condMax - condMin) * 255);
  brightness = constrain(brightness, 0, 255);
  analogWrite(ledPin, brightness);

  Serial.print("raw: "); Serial.print(raw);
  Serial.print("\tconductance: "); Serial.print(conductance, 6);
  Serial.print("\tbrightness: "); Serial.println(brightness);

  delay(50);
}
```



Placement order (XIAO SAMD21, no-rail mini breadboard):

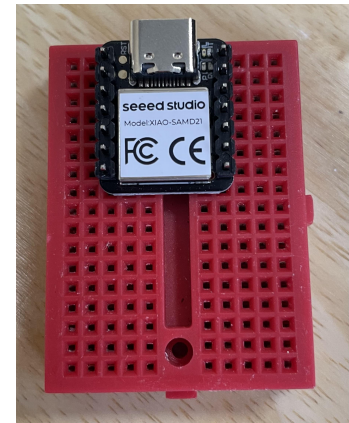
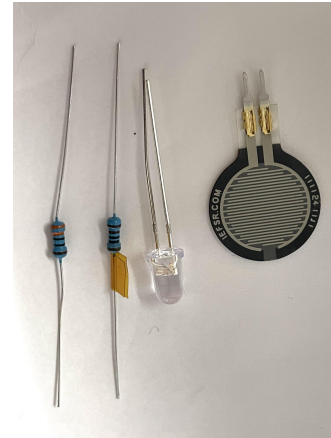
FSR circuit (chip's left side = A0)

1. Jumper wire from **3.3V-OUT** pin (chip's right side) → empty row **B**
2. **FSR402 leg 1** → row B (same row as the 3.3V jumper)
3. **FSR402 leg 2** → empty row **C**
4. **1k Ω resistor leg 1** → row C (same row as FSR leg 2 — this is the voltage-divider junction)
5. **1k Ω resistor leg 2** → empty row **D**
6. Jumper wire from a **GND** pin (chip's right side) → row D (same row as resistor leg 2)
7. Jumper wire from **A0** pin (chip's left side) → **row C** (the FSR–resistor junction — NOT row B)

LED circuit (chip's right side = D9)

8. Jumper wire from **D9** pin → empty row **E**
9. **330 Ω resistor leg 1** → row E (same row as the D9 jumper)
10. **330 Ω resistor leg 2** → empty row **F**
11. **LED anode (long leg)** → row F (same row as resistor leg 2)
12. **LED cathode (short leg)** → empty row **G**
13. Jumper wire from a **GND** pin → row G (same row as LED cathode)

Key check: row C must contain exactly three things — FSR leg 2, 1k Ω resistor leg 1, and the A0 jumper. That's the voltage-divider tap point that was wrong before. Row B (power) and row G (LED ground) each need their own separate GND/3.3V jumper since this board has no shared rails.

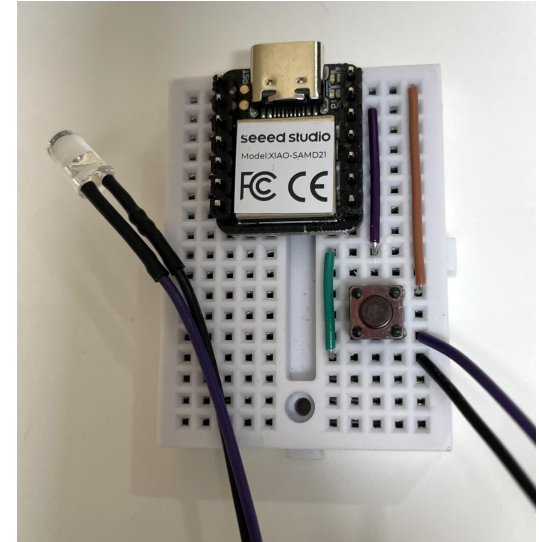
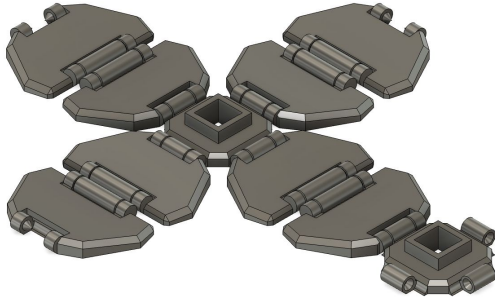


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Aug 31 - Modular Improvement 2

Module refinement and

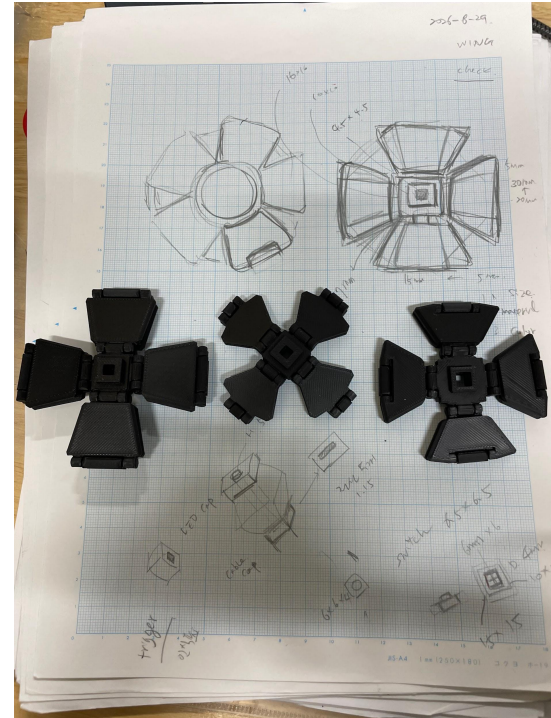
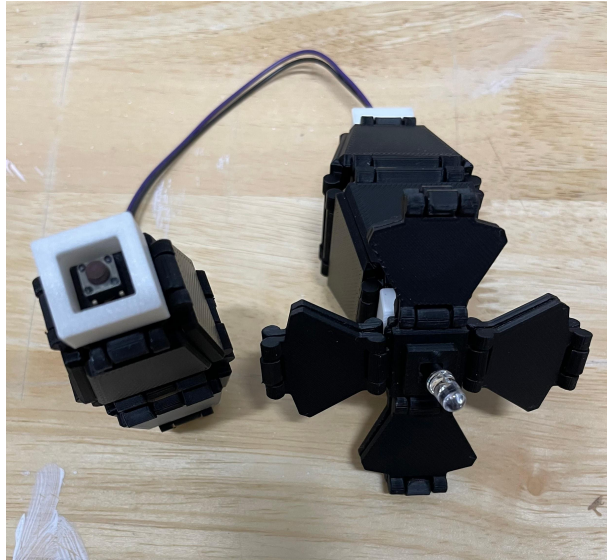
LED cable wiring



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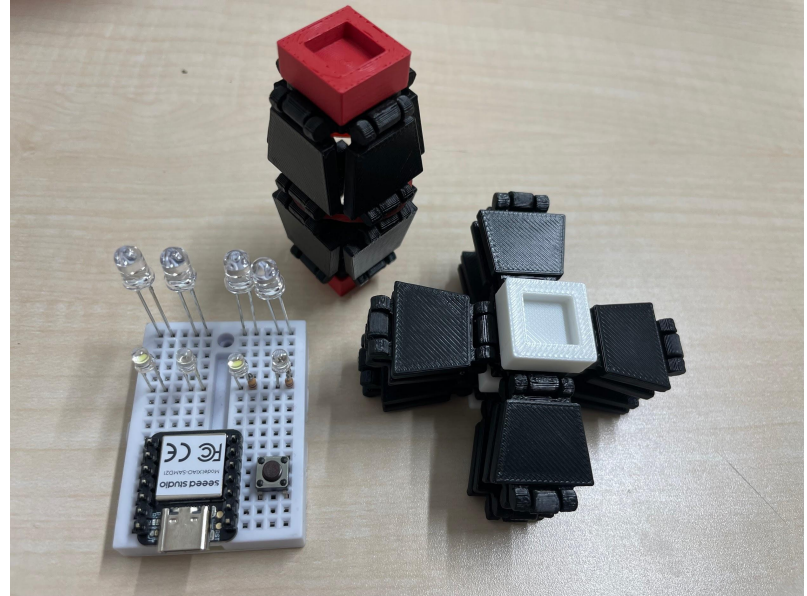
Aug 29, 30 - Modular Improvement 1

LED, Button wiring and Wing variations



Aug 28 - Circuit Build

Under time constraints,
I implement button and LED features.



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Aug 27 - Circuit Plan

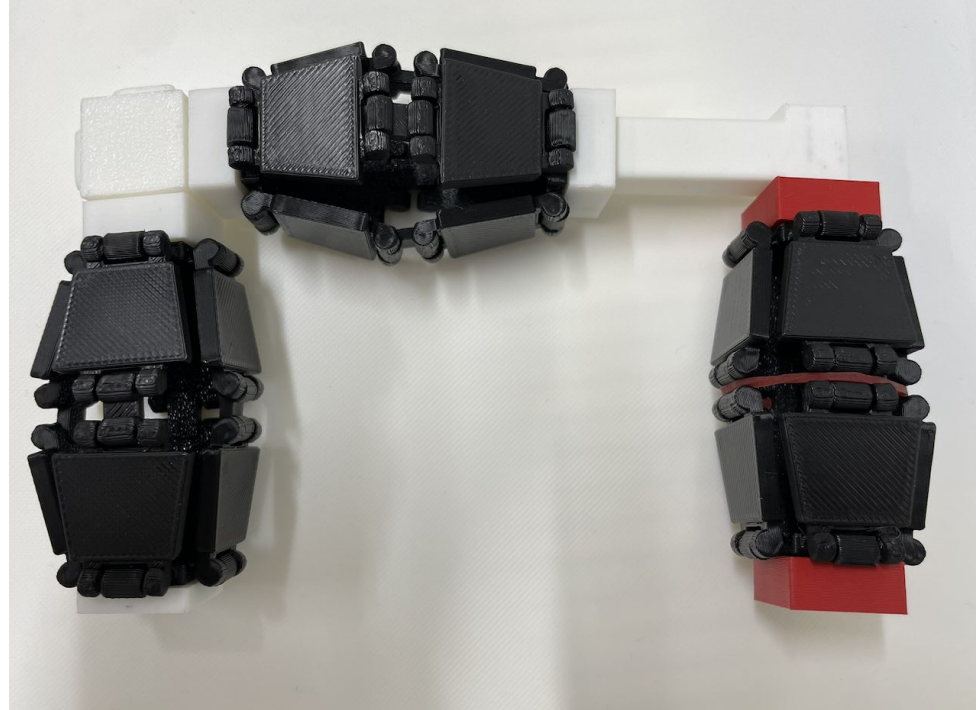
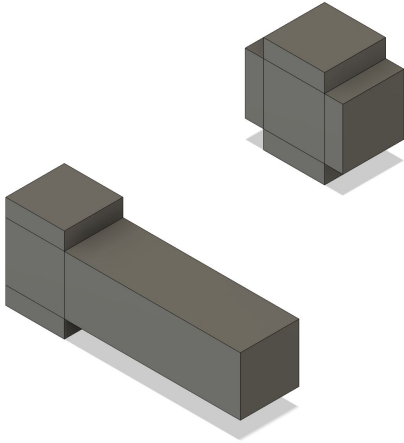
MCU	Seeed XIAO SAMD21
Sensor	Pressure, Photo diaod
Feedback	LED, Motor



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Aug 26 - Connector

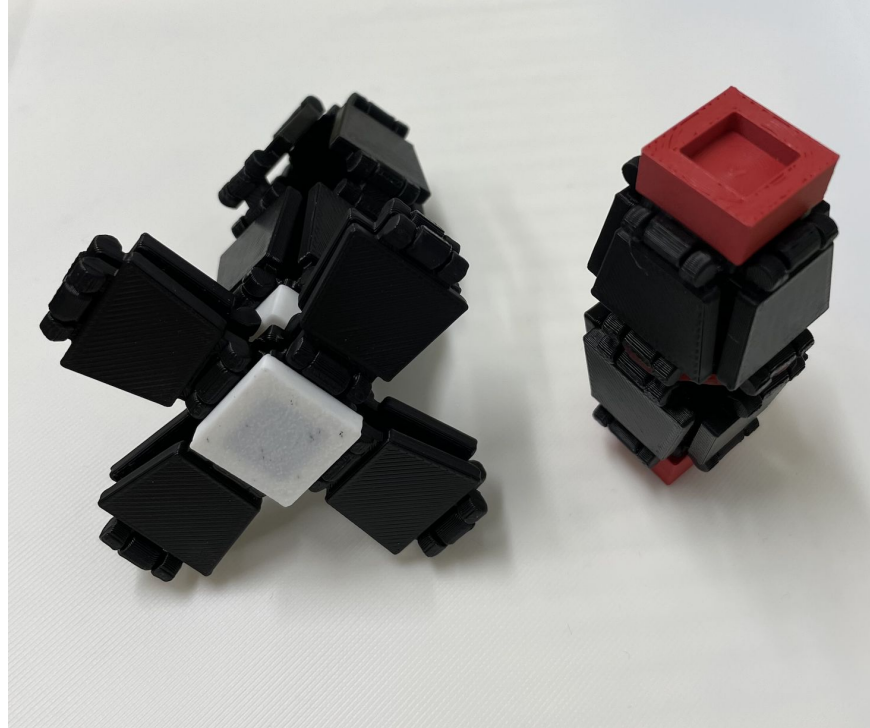
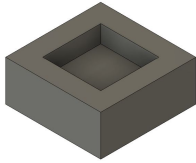
L - type connector long/short



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Aug 25 - MFT26 Layout

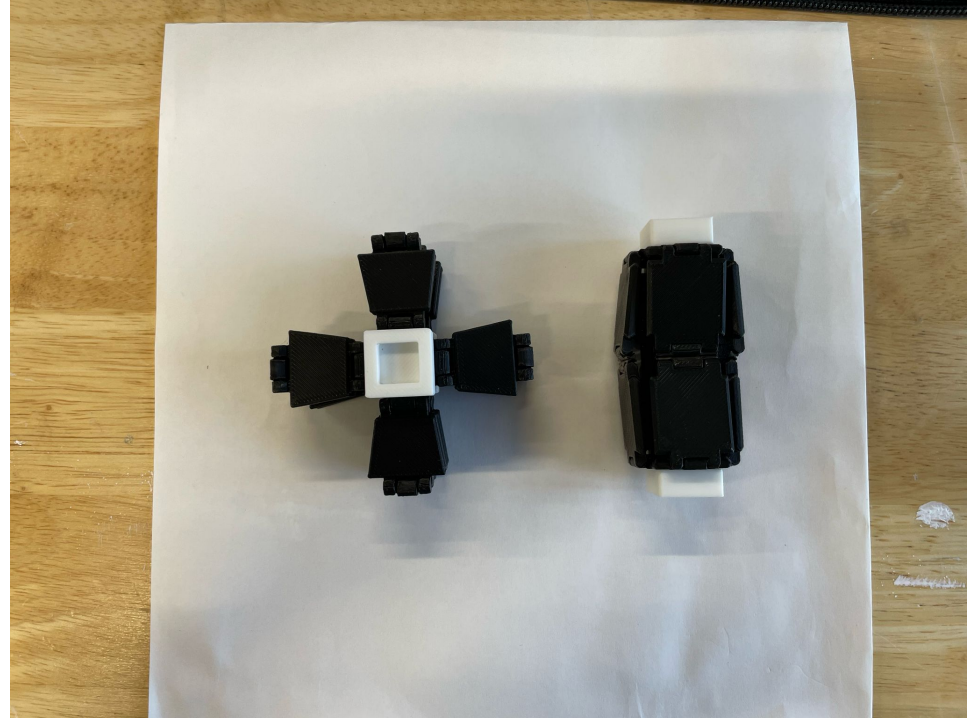
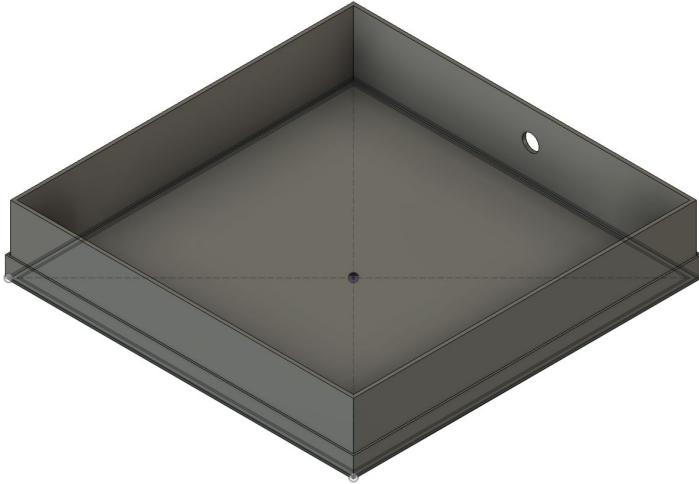
Form factor Variation



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Aug 24 - MFT26 Layout

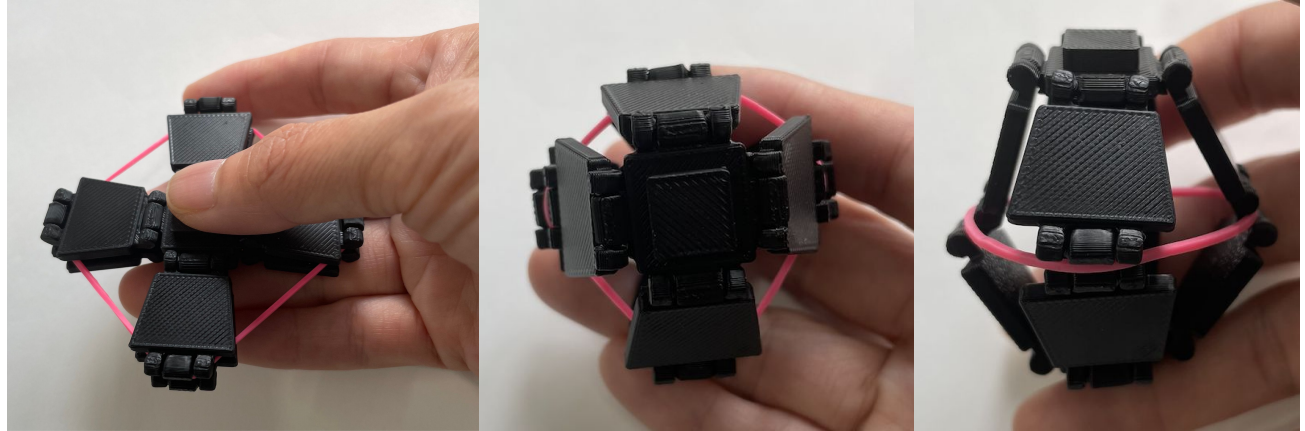
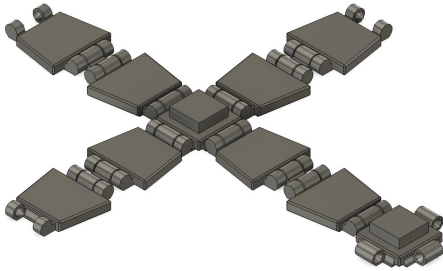
Maker Fair 2026 Exhibition Plan



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Aug 22 - Quadra trapezoid

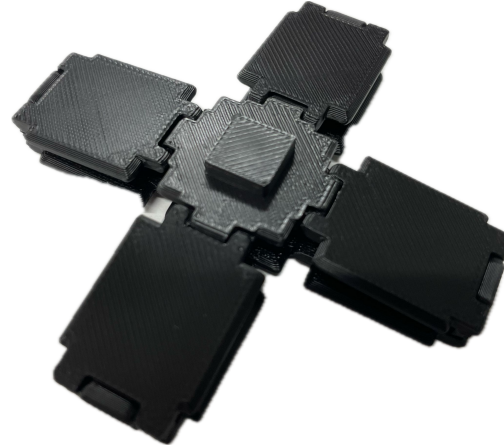
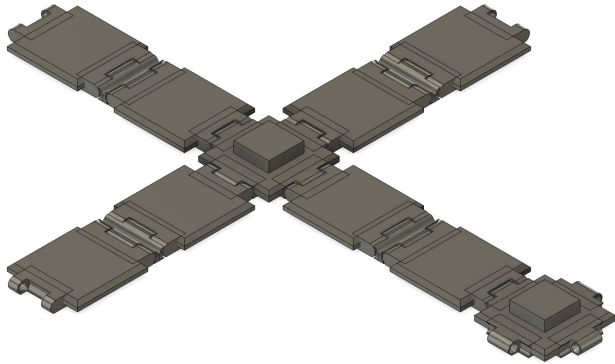
Trapezoid four wings with rubber band



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Aug 21 - Quadra perpendicular

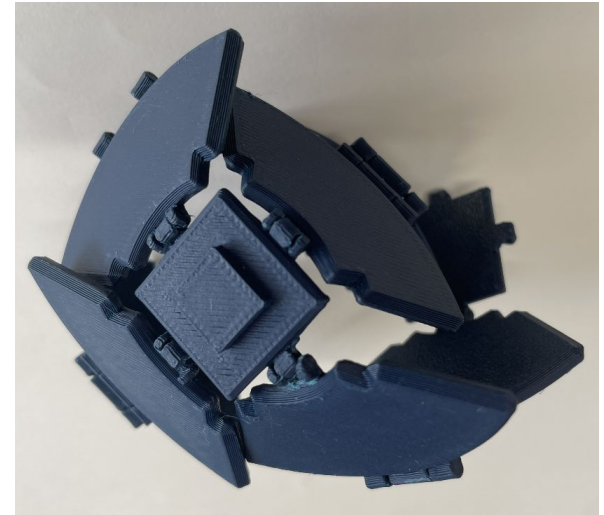
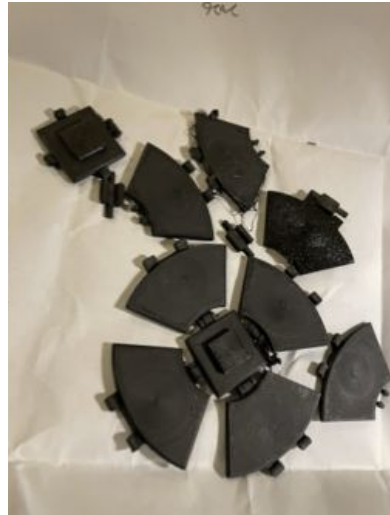
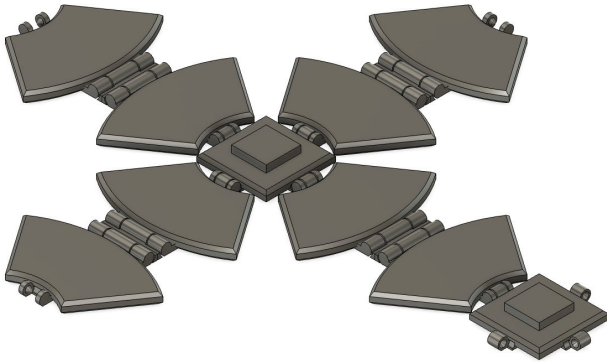
Perpendicular four wings modular



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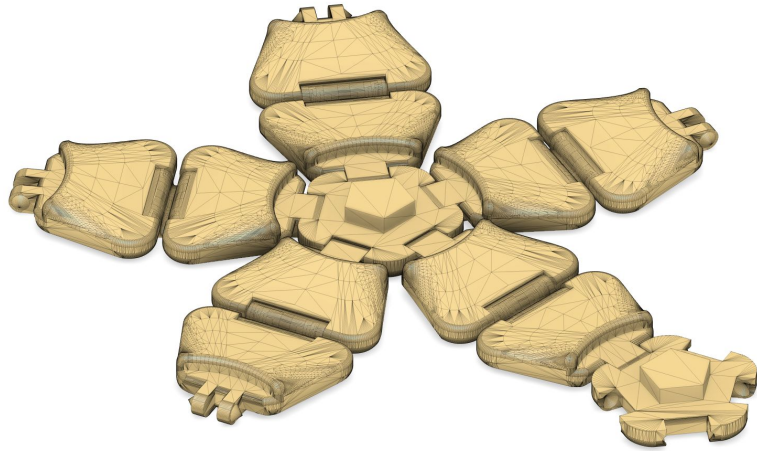
July 25 - Quadra

Initial four wings modular

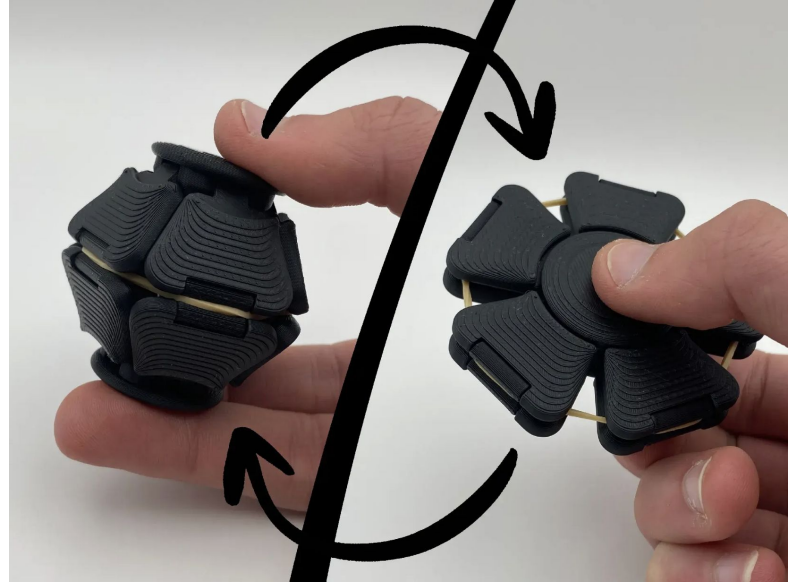


May 25 - FidgetSqueezer

Reference Model



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FUTURE CRAFT REAL PROJECT Maker Fair Tokyo 2026

Kyunghee YOO
2026 Summer